

SupGen

Verified reasoning data for the problems LLMs can’t solve

HIGHER ORDER COMPANY

Labs throw away the problems their models can’t solve.

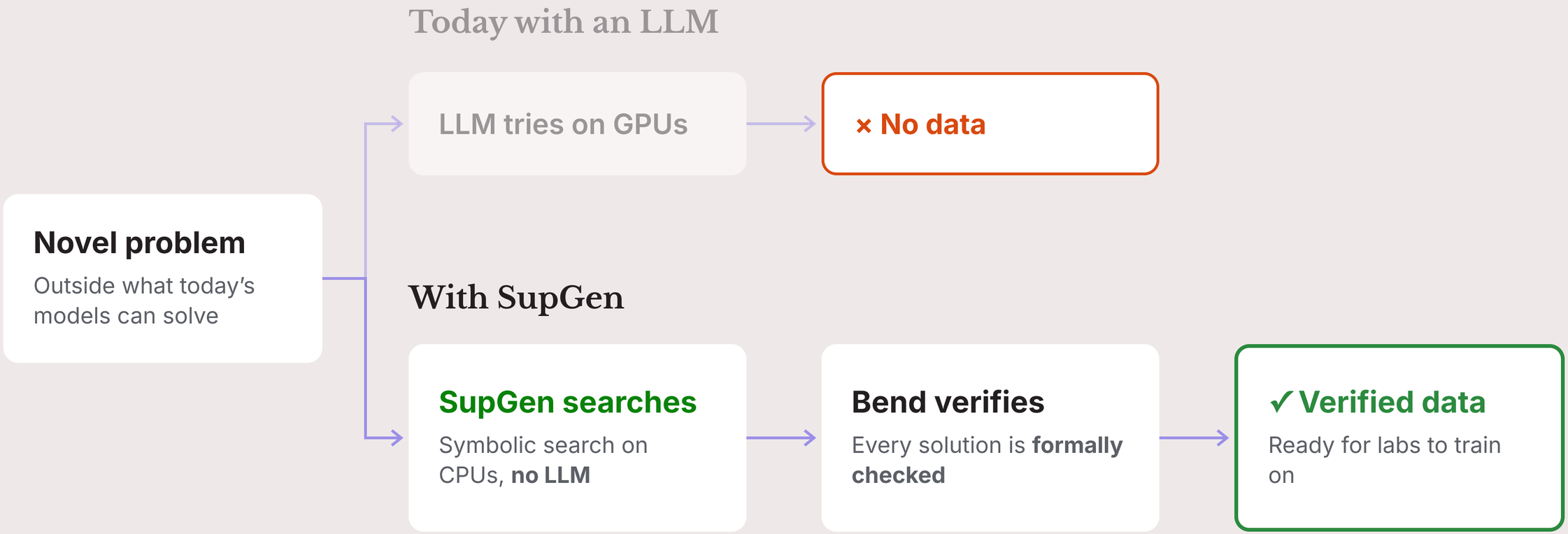
If the model can’t solve a problem, training learns nothing from it. So labs filter those problems out.



$95M **a year, if just 1% of one lab’s RL compute is wasted this way**

Data: Off-Context GRPO (2026), 64 attempts each on 595 problems. Illustrative cost: ~$19B OpenAI R&D compute × ~50% on RL × 1%.

SupGen turns those problems into verified data.



Most of the engine already exists.

Already have

HVM4
Parallel runtime that runs the search

Bend 2
Our language and proof verifier

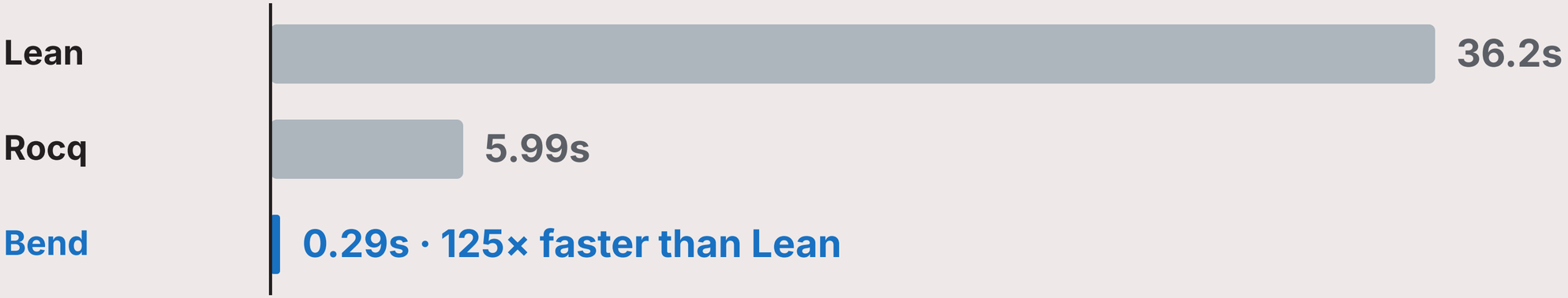
512 nodes
Cluster ready to run the search

What we'll build

SupGen Symbolic proof and program synthesis. This round funds it.

Bend checks proofs up to 125× faster than Lean

Time to check 12,800 definitions · Apple M4 Max · lower is better



Isabelle and Agda: over 5 minutes

0.83s

3,200 proofs · Rocq 9.53s, Lean over 5 min

0.38s

3,200 generic instantiations · Lean 19.2s

0.61s

400 smalltt trees · Lean 1.33s

Benchmarks published by the Bend team in the Bend repository (github.com/bendlang/bend)

Developers already back what we build

24.9k
GitHub stars on Bend

6.9k
members in our Discord

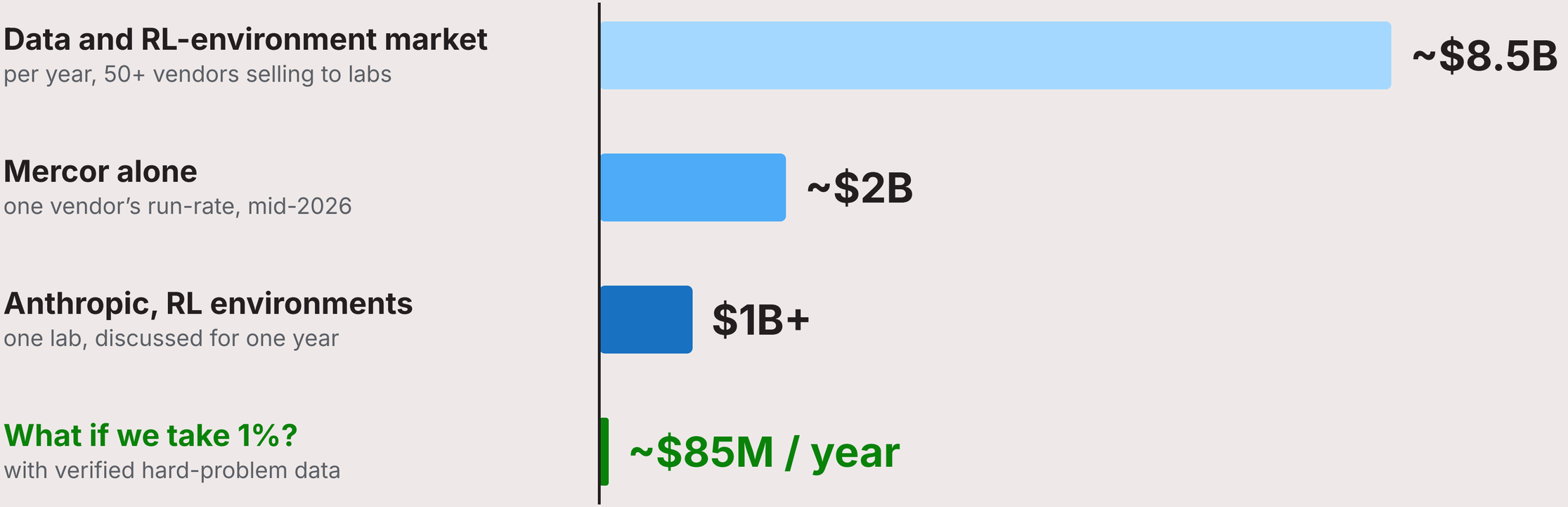
1,041
Hacker News points at Bend's launch

$4M
seed raised in 2023

Bend 2 (Sep 2026) hit the Hacker News front page again: 616 points, 327 comments. HVM1 and Kind add another 15k GitHub stars.

As of Nov 7, 2026. Sources: GitHub, Discord, Hacker News; seed amount from Victor Taelin's 2024 post.

Labs already pay billions for training data



Sources: industry tally, Jul 2026 (rough); Mercor H1 2026 revenue $614M; The Information, Sep 2025 (discussion, not a budget). 1% is a scenario, not a forecast.

We sell the data labs can’t make

FIRST **Verified hard-problem datasets for AI labs**

NEXT **Reasoning API: send a problem, get a proof**

LLM training runs on GPUs that cost a fortune. SupGen runs on CPUs, so every verified solution costs a fraction.

We are raising $8M

SAFE · $120M post-money cap · $100k minimum · all to R&D

1

SupGen beyond simple types

2

First dataset on problems frontier models fail

3

Paid pilot with one AI lab

High-risk research investment · investors@higherorderco.com

HOC Team

Victor Taelin

Founder, CEO · created HVM, Bend and Kind

Nicolas Abril

Researcher

Paulo Cavalcanti

Researcher

Vitor Chiarelli Neves

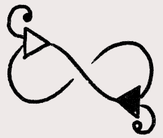
Cofounder

Lorenzo Battistela

Researcher

The bitter lesson, taken to the extreme

The simplest method plus massive compute wins. For reasoning, that method is proof search. We’re building it.



Sources

Problem: zero learning signal on unsolved problems

- Advantage Collapse in Group Relative Policy Optimization (arXiv, 2026)
- Off-Context GRPO: Learning to Reason on Hard Problems (arXiv, 2026)
- QuestA: Expanding Reasoning Capacity via Question Augmentation (arXiv)
- LatentRevise: Learning from Zero-Hit Reasoning (arXiv, 2026)
- Interconnects: DAPO and Open-Reasoner-Zero filter zero-pass problems
- OpenAI 2026 R&D compute ~$19B, Epoch AI estimate (via HeroHunt)
- RL ≈ half of frontier-lab compute, SemiAnalysis estimate (via Exobrain)

Market: what labs pay for data

- AI training-data sector revenue (~$8.5B), BigGo Finance
- AI training-data market concentration, 50+ vendors (Pebblous)
- Mercor H1 2026 revenue $614M, BigGo Finance
- Anthropic discussed $1B+ on RL environments (TechCrunch, citing The Information)
- Epoch AI: an FAQ on RL environments

Engine: HVM4, SupGen, Bend 2

- Victor Taelin: SupGen built into HVM4
- Bend 2 release (Akita on Rails, Sep 2026)

Traction and Bend benchmarks

- Bend repository: GitHub stars and checker benchmarks (README)
- Bend Discord
- Hacker News: Bend launch, May 2024 (1,041 points)
- Hacker News: Bend 2 launch, Sep 2026 (616 points)
- Victor Taelin: HOC raised a $4M seed round in 2023